Enron Email Corpus – Week 5

Cleaning the corpus, reworking the review

Data Mining Lab SoSe 26 – Task 5

June 3, 2026

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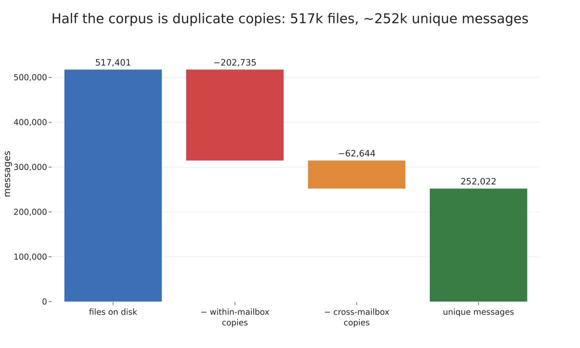
From raw files to a clean corpus

Last week's raw-file problems: about half were duplicates, bodies carried quoted history, addresses were split per person, and there were no thread headers. We cleaned the corpus in four passes into `eda-5/clean/`:

- **Deduplicate** to one row per distinct message
- **Clean bodies**: strip quoted history, HTML, disclaimers
- **Resolve addresses**: spelling variants to one person
- **Reconstruct threads**: subject + participant, 30-day gap

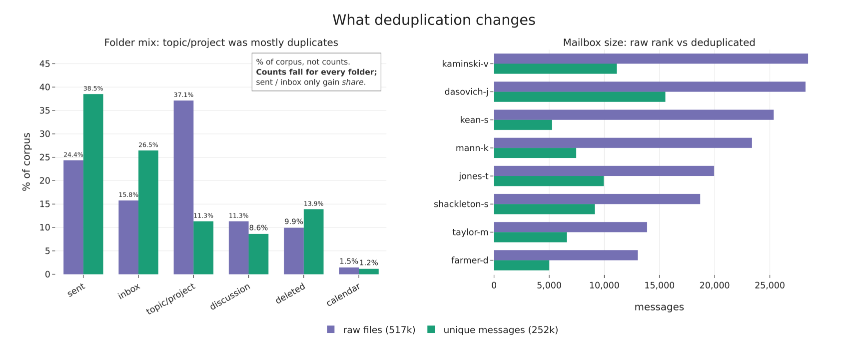
Result: **252,022** messages and **1.37 M** recipient rows.

Half the files are duplicates



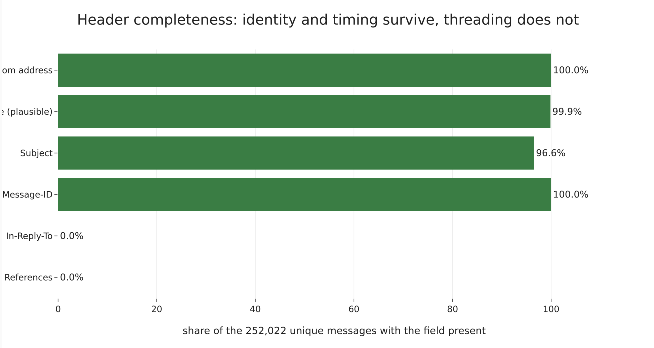
517,401 files collapse to **252,022** distinct messages (**51%** duplication). Every later count is on the distinct messages.

What dedup changes



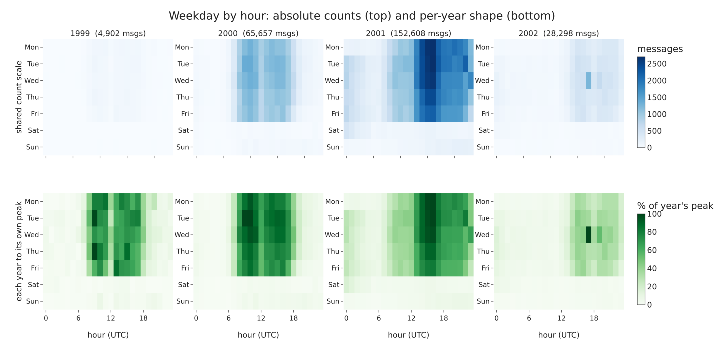
Two week-2 numbers were artifacts. `topic/project` folders drop from **37%** to about **11%**; the largest mailbox flips from `kaminski-v` to `dasovich-j`. Left bars are *share* of the corpus, not counts.

How complete is the data



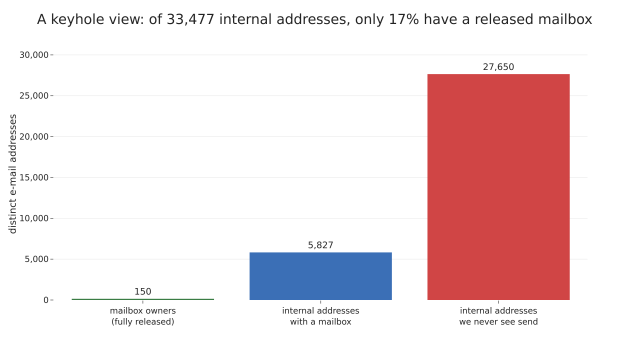
Identity and timing are near-complete: sender **100%**, timestamp **99.9%**, subject **96%**. The threading headers are gone.

The 24-hour activity plot, fixed



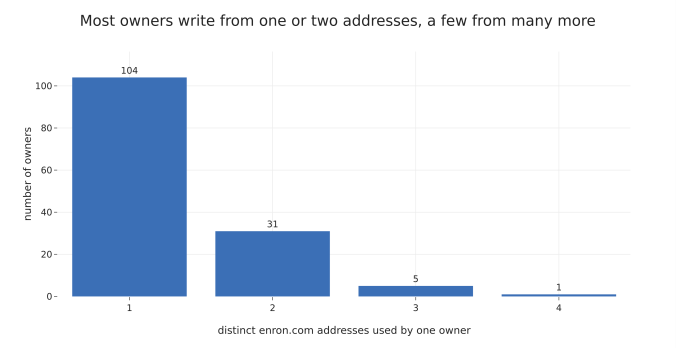
Top row (blue): one shared count scale, honest about volume. Bottom row (green): each year on its own scale, honest about the daily shape. Empty cells are white.

A keyhole onto Enron



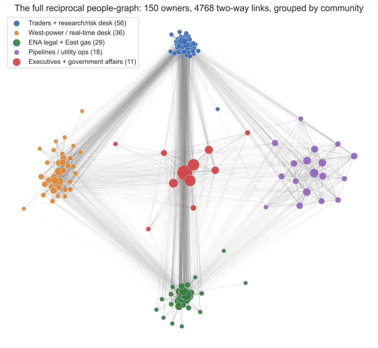
The 150 mailboxes wrote to about **33,000** distinct internal accounts, but only about **17%** of those have a mailbox of their own in the release.

Addresses are not people



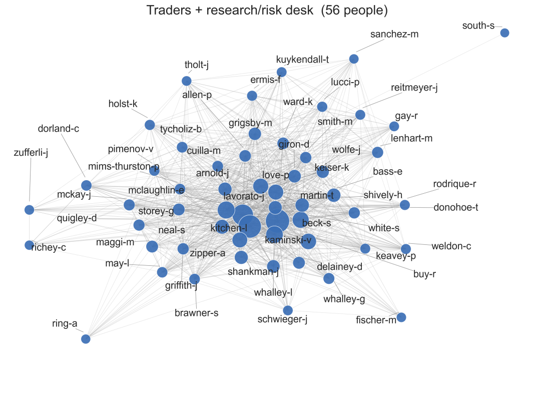
Of 150 owners, **104** write from one address, **31** from two, **6** from three or four. The multi-address cases are messy formatting of one name, not extra accounts.

The reciprocal graph, grouped



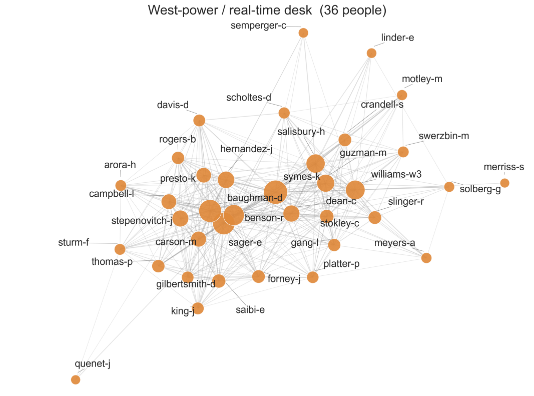
150 owners, **4,768** two-way links, coloured by a modularity split. Dense, but not structureless: five groups separate out.

Working group 1: trading floor + research



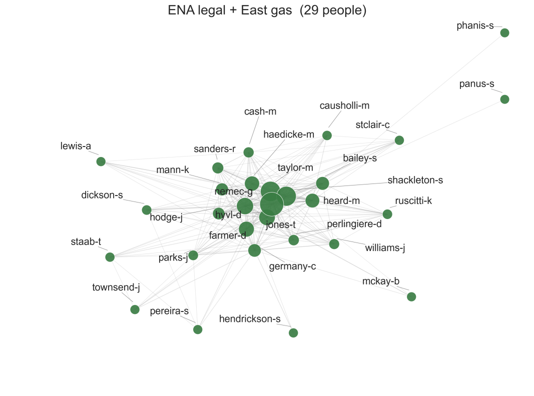
The largest community (56): the trading desk, research and risk around it.

Working group 2: West-power / real-time desk



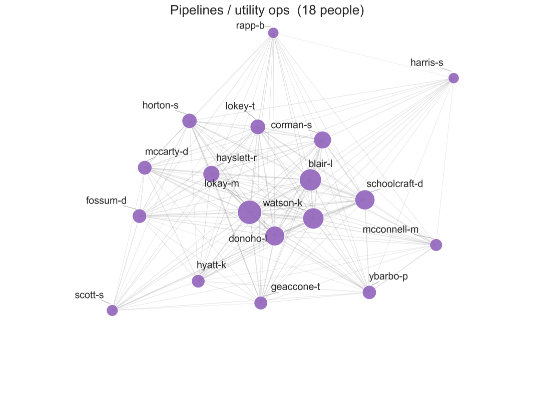
36 people: the West-power / real-time desk that ran the California book.

Working group 3: ENA legal + East gas



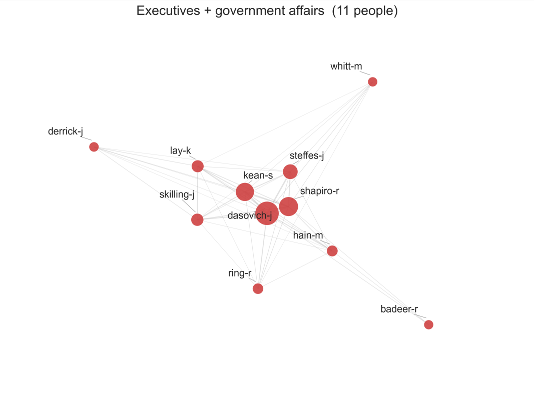
29 people: the ENA legal department and the East gas desk it served.

Working group 4: pipelines / utility ops



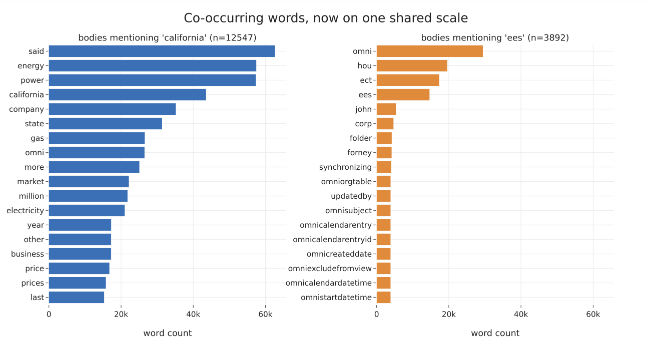
18 people: the pipelines and utility-operations group.

Working group 5: executives + government affairs



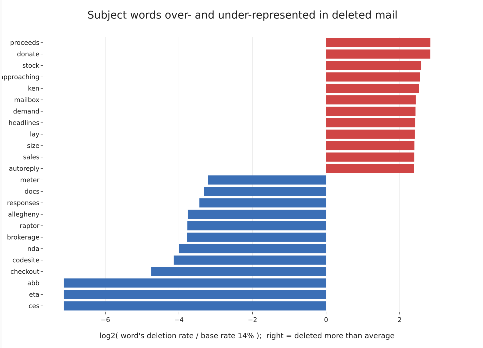
The smallest (11), and the core that ties the others together.

California vs EES, on one scale



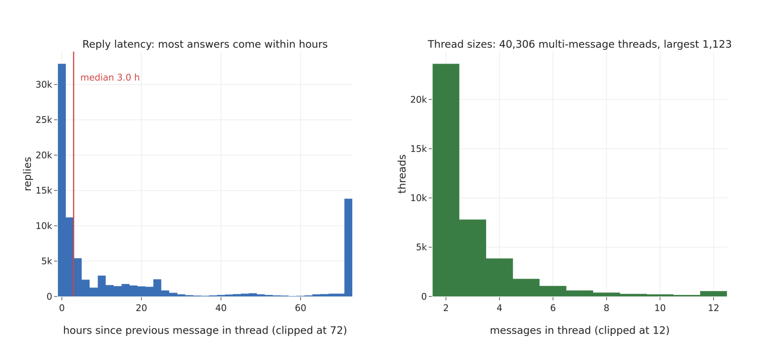
Now on the full corpus, not a 25k sample: about **12,500** bodies mention California against about **3,900** for EES. Same axis, so the gap is real.

What gets deleted



Deletion is topic-driven. Base rate about **14%**. Far above it sit one protest campaign (`donate`, `stock`, `lay`) and machine noise; business words are kept.

Conversations, finally



Reconstruction gives **40,306** threads. Replies are fast: median about **3 hours**, three quarters inside a day. Sizes are heavily skewed, mostly short.

Where we stand

The corpus is now deduplicated, body-cleaned, address-resolved and thread-aware: **252,022** messages from 150 mailboxes, 1999–2002.

Three week-2 numbers moved once we counted messages, not files: the folder mix inverts, the largest mailbox re-ranks, and the social core is denser than it looked. The new material is threads, fast replies and a heavy-tailed size distribution.

The standing caveat is coverage, and the reconstructed threads are a heuristic standing in for headers that no longer exist.

Questions?

Full write-up: `eda-5/DATASET_CLEANING.md` and `eda-5/enron_eda_week4.ipynb`